Task 1

Multimodel List of Groq

Groq multimodal models

GroqCloud

S.No	Modal name	Modalities	Key capabilities
1	Llama 4 Scout	Text + Images → Text	Image understanding (vision)
2	Llama Guard 4	Text + Images → Text	• Content moderation • Safety classification (harmful content detection)
3	Whisper	Audio → Text	• Speech-to-text • Transcription
4	Orpheus TTS	Text → Audio	• Text-to-speech • Natural voice generation

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OpenAI multimodal models

OpenAI

S.No	Model name	Features	Key capabilities
1	GPT-5.4 Flagship model	Text + Images → Text	• Best intelligence for agentic & coding workflows • Vision & document understanding • Computer use support • 1M token context window
2	GPT-5.4 Pro Premium variant	Text + Images → Text	• Smarter & more precise responses • Complex professional workflows • Advanced reasoning & analysis • Vision & document processing
3	GPT-5.4 Mini Efficient variant	Text + Images → Text	• Strongest mini model for coding & subagents • Low latency & cost-optimized • Computer use capable • Vision support
4	GPT-4o Omni model	Text + Images + Audio → Text + Audio	• Native end-to-end audio understanding • Real-time voice interaction (~320ms latency) • Multilingual text, vision & audio • Best voice model for audio in / audio out
5	GPT Image 1.5 Image generation	Text + Images → Images	• High-fidelity image generation & editing • Instruction-following for image tasks • Used in ChatGPT image generation • Supports image input for editing
6	Sora Video generation	Text + Images → Video	• Text-to-video generation • Image-to-video animation • Short-form video prototyping • Creative & marketing use cases
7	Whisper Speech-to-text	Audio → Text	• Speech recognition & transcription • Multilingual audio support • Realtime transcription via Realtime API • High accuracy across accents
8	TTS (Text-to-Speech) Audio generation	Text → Audio	• Natural-sounding voice synthesis • Multiple voice styles available • Low-latency audio output • Voice agent support

Task 2

Take 3 models from each service provider and provide the Context window, Input and Output token Cost

Service Provider	Model	Context Window	Input Token Cost	Output Token Cost
Open AI	GPT-5.4 Pro	1.05 M	$30 /M	$180/M
	GPT-5.3 Codex	400K	$1.75/M	$14/M
	GPT-5.4	1.05 M	$2.50/M	$15/M
X AI	Grok 4.20 Multi-Agent	2 M	$ 2/M	$6/M
	Grok Code Fast 1	256K	$0.20/M	$1.50/M
	Grok 3	131K	$3/M	$15/M
Google	Gemini 3.1 Flash Lite Preview	1.05M	$0.25/M	$1.50/M
	Nano Banana 2	131K	$0.50/M	$3/M
	Gemini 3.1 Pro	1.05 M	$2/M	$12/M
Anthropic	Claude Sonnet 4.6	1M	$3/M	$15/M
	Claude Opus 4.6	1M	$5/M	$25/M
	Claude 3 Haiku	200K	$0.25/M	$1.25/M
Alibaba	Qwen 3.5 -27B	256K	$0.195/M	$1.56/M
	Qwen 3.5 Flash	1M	$0.065/M	$0.26/M
	Qwen 3 Max Thinking	262K	$0.78/M	$3.90/M
Meta	Llama 3.1 8B Instruct	16K	$0.02/M	$0.05/M
	Llama 3.3 70B Instruct	131K	$0.10/M	$0.32/M
	Llama 4 Maverick	1.05M	$0.15/M	$0.60/M

Task 3

Use the Tokenizer with different prompt and see how the tokens are generated

Prompt 1

Explain how black holes form and what happens to matter that crosses the event horizon

Prompt 2

Give me a quick recipe for a creamy pasta dish using ingredients commonly found at home

Prompt 3

What are the key differences between machine learning and deep learning, and when should each be used?

Prompt 4

What is the Capital of France?